

Context-Aware Student Digital Twin for Personalized Microlearning

Summary

This project develops a student-controlled, context-aware digital twin platform for higher education. The platform converts short, otherwise underused periods outside formal teaching—commute segments, inter-class gaps, waiting periods, quiet breaks, and other low-friction moments—into safe, pedagogically meaningful 3–15 minute learning opportunities. At suitable moments, it delivers brief, course-aligned interventions such as quizzes, flashcards, reading snippets, infographics, or short audio recaps designed to support retrieval practice, spaced review, concept reinforcement, and measurable learning gain.

The platform is designed as a bidirectional digital twin system with two equally important outputs. First, a student-facing adaptive microlearning loop helps each learner study in short, well-timed moments. Second, an educator-facing course and cohort intelligence layer helps lecturers, professors, advisors, learning designers, and program leaders see what students know, where they struggle, where overload is emerging, and how course design can be adjusted before summative assessment pressure peaks.

The central scientific contribution is an event-centric architecture positioned between multimodal sensing and adaptive educational decision-making. Rather than operating directly on heterogeneous raw signals, the system transforms smartphone, institutional, timetable, LMS, and optional external inputs into probabilistic, time-bounded, semantically structured contextual events such as `WaitingWindow`, `CommuteAudioWindow`, `TransitIdle`, `FocusedReviewWindow`, `OverloadRiskWindow`, and `LowReceptivityWindow`. These events enrich a persistent Student Twin with explicit confidence, provenance, temporal validity, modality relevance, and safety status.

This architecture allows the system to model not only what a student should learn next, but also whether the current moment is suitable for intervention, which modality fits the context, how long the opportunity is likely to last, and how certain the system is that a learning action will be useful. Student outcomes then flow back into a Course/Cohort Twin that aggregates permissioned evidence into educator-facing insights: mastery distributions, misconception clusters, pacing strain, overload windows, content units associated with high cognitive burden, and exam-readiness indicators.

The platform is explicitly designed for device-only, device-edge, and device-edge-cloud operation. Its logical behavior remains consistent across modes, while computation is partitioned according to privacy constraints, latency budgets, energy limits, legal permissions, infrastructure maturity, and institutional governance requirements. Raw high-frequency sensor data remain local wherever possible; compact features, contextual events, policy logs, and pseudonymous learning-state summaries are used for central processing and analytics.

The intended outcome is a privacy-aware, standards-based, auditable, and scalable digital twin platform for adaptive learning and teaching intelligence. The project will deliver formal event and twin schemas, a curriculum alignment graph, reference implementations, LMS integration,